

C PROGRAMMING :FIRST FIT

```
GNU nano 9.0 ff.c *
#include <stdio.h>
int main(){
    int blockSize[20], processSize[20];
    int allocation[20];
    int nb, np, i, j;
    printf("Enter Number of Blocks: ");scanf("%d",&nb);
    printf("Enter Number of Processes: ");scanf("%d",&np);
    printf("Enter Block Sizes:\n");
    for(i=0;i<nb;i++){
        scanf("%d",&blockSize[i]);
        printf("Enter Process Sizes:\n");
        for(i=0;i<np;i++){
            scanf("%d",&processSize[i]);
            for(i=0;i<nb;i++){
                allocation[i]=-1;
                for(j=0;j<np;j++){
                    if(blockSize[j] >= processSize[i]){
                        allocation[i]=j;
                        blockSize[j]-=processSize[i];
                        break;
                    }
                }
                printf("\nProcess No\tProcess Size\tBlock No\n");
                for(i=0;i<np;i++){
                    printf("%d\t%d\t%d\t",i+1,processSize[i]);
                    if(allocation[i]!=-1)
                        printf("%d\n",allocation[i]+1);
                    else
                        printf("Not Allocated\n");
                }
            }
        }
    }
    return 0;
}
```

OUTPUT :

```
zsh: current history file /home/kali/.zsh_history
(kali@thamim72~)
$ nano ff.c
(kali@thamim72~)
$ gcc ff.c -o ff
(kali@thamim72~)
$ ./ff
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
2
2
Enter Process Sizes:
2
2
Process No      Process Size    Block No
1               2               1
2               2               2
(kali@YUVARAJK)-[~]
$
```

SHELL PROGRAMMING : FIRST FIT

The screenshot shows a Kali Linux terminal window with the following content:

```

kali@YUVARAJK: ~
Session Actions Edit View Help
GNU nano 9.0 ff.sh
#!/bin/bash
echo "First Fit Memory Allocation Demonstration"

62
blocks=(100 500 200 300 600)
processes=(212 417 112 426)
echo "Memory Blocks: ${blocks[@]}"
echo "Processes: ${processes[@]}"
echo "Allocation Performed Using First Fit"

page.sh
Enter Page Count:
Enter Page Number for Page 0:
Enter Page Number for Page 1:
Enter Page Number for Page 2:
Enter Page Number for Page 3:
Enter Logical Address:
Page Number: 1
Enter File:
Enter Size:
Enter Address:

Read 10 lines
H Help W Write Out W Where Is C Cut E Execute L Location M-U Undo M-A Set Mark M-T To Bracket M-P Previous
X Exit R Read File A Replace U Paste J Justify G Go To Line E Redo M-E M-6 Copy B Where Was M-B Next

```

OUTPUT :

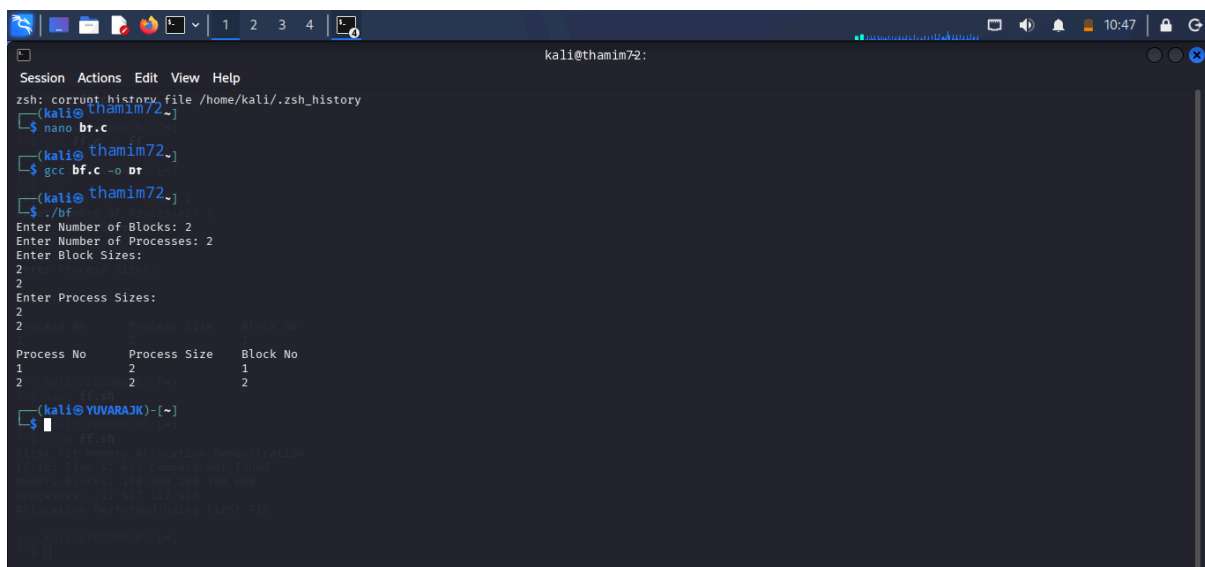
```
kali@thamim72:~$ nano ff.c
Session Actions Edit View Help
└─$ nano ff.c
(kali@thamim72~)$ gcc ff.c -o ff
(kali@thamim72~)$ ./ff
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
2
2
Enter Process Sizes:
2
2
2
Process No      Process Size    Block No
1                2                1
2                2                2
(kali@thamim72~)$ nano fr.sh
(kali@thamim72~)$ bash fr.sh
First Fit Memory Allocation Demonstration
ff.sh: line 4: 62: command not found
Memory Blocks: 100 500 200 300 600
Processes: 212 417 112 426
Allocation Performed Using First Fit
(kali@thamim72~)$
```

C PROGRAMMING : BEST FIT



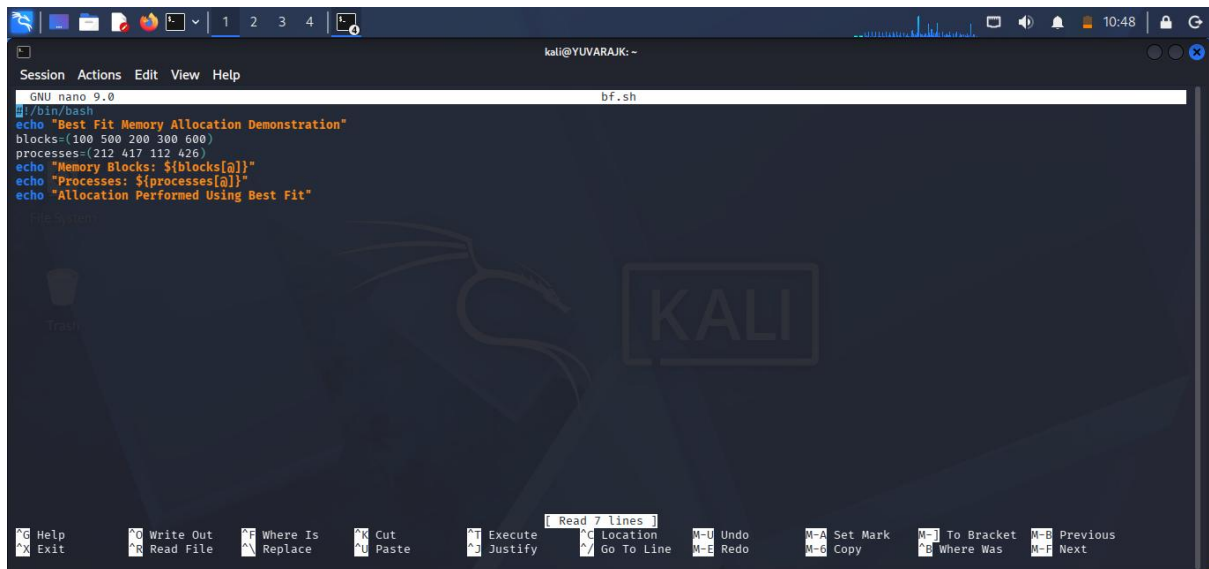
```
GNU nano 9.0 bf.c *
#include <stdio.h>
int main()
{
    int blockSize[20], processSize[20];
    int allocation[20];
    int nb,np,i,j,bestIdx;
    printf("Enter Number of Blocks: ");scanf("%d",&nb);
    printf("Enter Number of Processes: ");scanf("%d",&np);
    printf("Enter Block Sizes:\n");
    for(i=0;i<nb;i++)
        scanf("%d",&blockSize[i]);printf("Enter Process Sizes:\n");
    for(i=0;i<np;i++)
        scanf("%d",&processSize[i]);
    for(i=0;i<np;i++){allocation[i]=-1;
        for(j=0;j<nb;j++){
            if(blockSize[j] >= processSize[i]){
                if(bestIdx== -1 || blockSize[j] < blockSize[bestIdx])bestIdx=j;
            }
        }
        if(bestIdx != -1){
            allocation[i]=bestIdx;
            blockSize[bestIdx]=processSize[i];
            printf("\nProcess No\tProcess Size\tBlock No\n");
            for(i=0;i<np;i++){
                printf("%d\t\t%d\t\t",i+1,processSize[i]);
                if(allocation[i] != -1)
                    printf("%d\n",allocation[i]+1);
                else
                    printf("Not Allocated\n");
            }
        }
    }
    return 0;
}
```

OUTPUT :



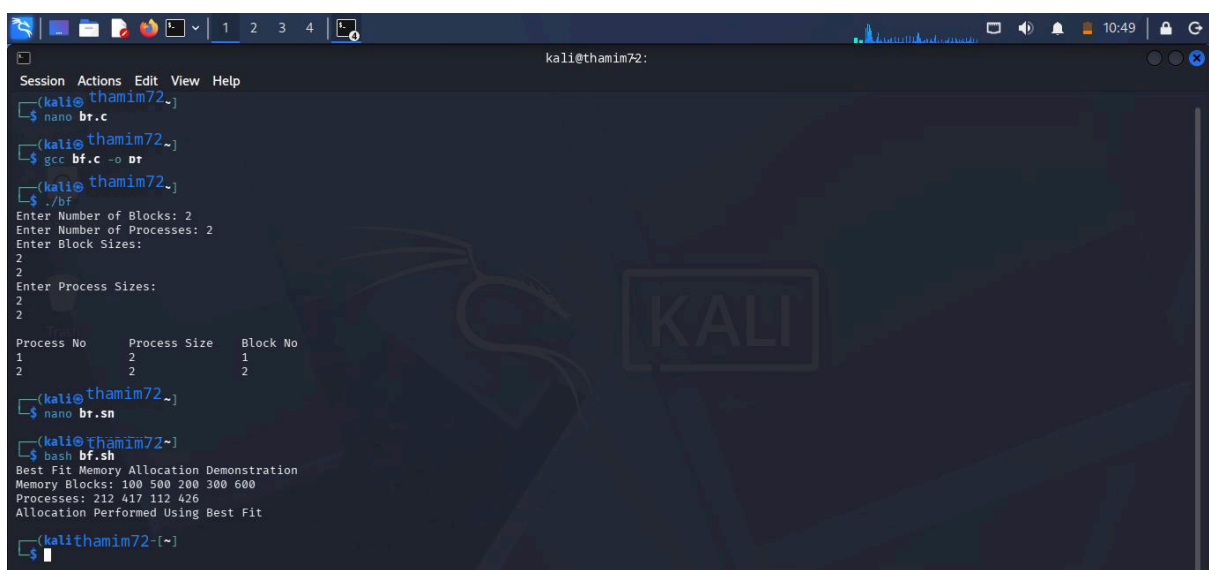
```
kali@thamim72:~$ zsh: corrupt history file /home/kali/.zsh_history
(kali@thamim72~)
(kali@thamim72~) nano bf.c
(kali@thamim72~)
(kali@thamim72~) gcc bf.c -o bf
(kali@thamim72~)
(kali@thamim72~) ./bf
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
2
2
Enter Process Sizes:
2
2
Process No      Process Size    Block No
1                2                1
2                2                2
(kali@YUVARAJK)-[~]
$
```

SHELL PROGRAMMING : BEST FIT



```
GNU nano 9.0 bf.sh
#!/bin/bash
echo "Best Fit Memory Allocation Demonstration"
blocks=(100 500 200 300 600)
processes=(212 417 112 426)
echo "Memory Blocks: ${blocks[@]}"
echo "Processes: ${processes[@]}"
echo "Allocation Performed Using Best Fit"
```

OUTPUT :



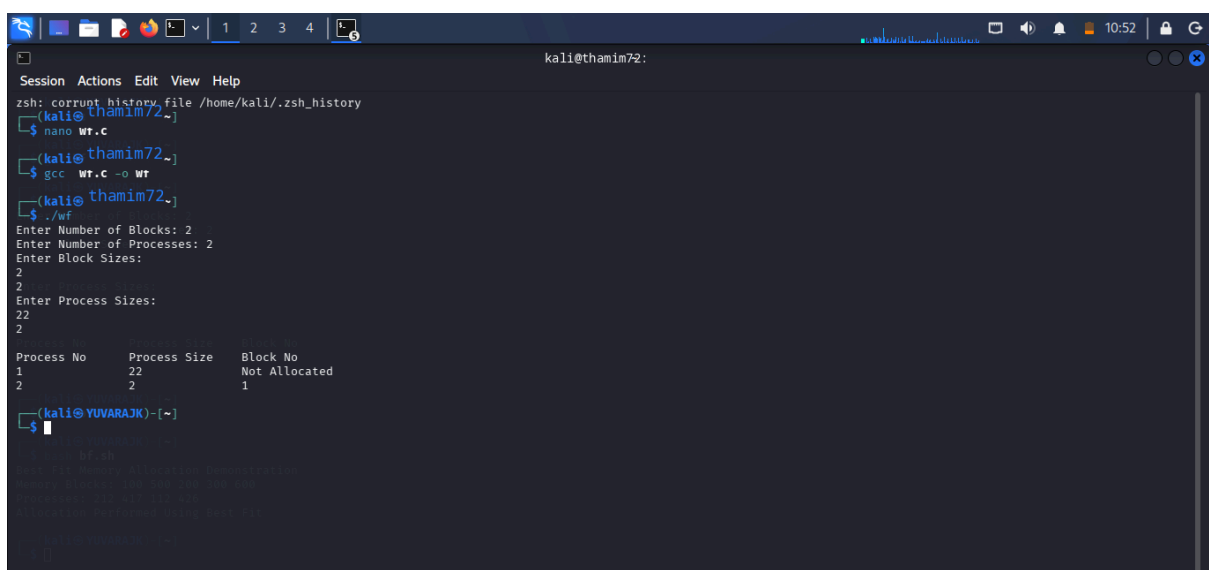
```
kali@thamim72:~$ nano br.c
kali@thamim72:~$ gcc bf.c -o br
kali@thamim72:~$ ./br
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
2
2
Enter Process Sizes:
2
2
Process No      Process Size    Block No
1                2                1
2                2                2
kali@thamim72:~$ nano br.sn
kali@thamim72:~$ bash bf.sh
Best Fit Memory Allocation Demonstration
Memory Blocks: 100 500 200 300 600
Processes: 212 417 112 426
Allocation Performed Using Best Fit
kali@thamim72:~$
```

C PROGRAMMING : WORST FIT



```
GNU nano 9.0 wf.c *
#include <stdio.h>
int main(){
    int blockSize[20], processSize[20];
    int allocation[20];
    int nb,np,i,j,worstIdx;
    printf("Enter Number of Blocks: ");scanf("%d",&nb);
    printf("Enter Number of Processes: ");scanf("%d",&np);
    printf("Enter Block Sizes:\n");
    for(i=0;icnb;i++){
        scanf("%d",&blockSize[i]);printf("Enter Process Sizes:\n");
    }
    for(i=0;icnp;i++){
        scanf("%d",&processSize[i]);
    }
    for(i=0;icnp;i++){
        allocation[i]=-1;
        for(j=0;jcnp;j++){
            if(blockSize[j] >= processSize[i]){
                if(worstIdx== -1 || blockSize[j] > blockSize[worstIdx])worstIdx=j;
            }
        }
        if(worstIdx!= -1){
            allocation[i]=worstIdx;
            blockSize[worstIdx]=processSize[i];
            printf("\nProcess No\tProcess Size\tBlock No\n");
            for(i=0;icnp;i++){
                printf("%d\t\t%d\t\t",i+1,processSize[i]);
            }
            if(allocation[i]!=-1){
                printf("%d\n",allocation[i]+1);
            }
            else{
                printf("Not Allocated\n");
            }
        }
    }
    return 0;
}
```

OUTPUT :

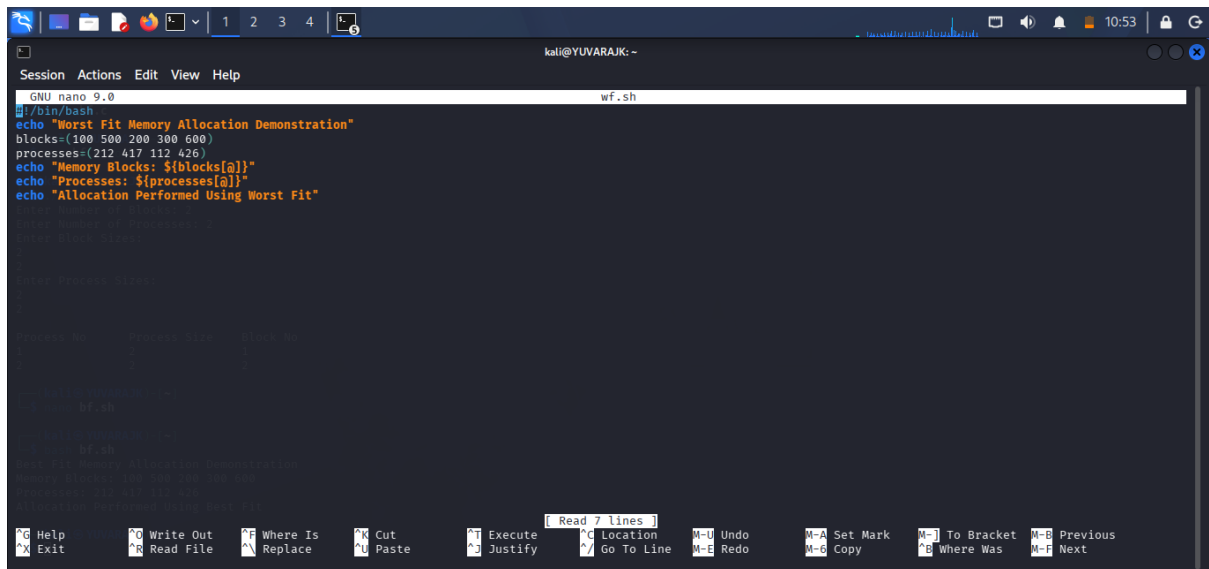


```
kali@thamim72:~$ zsh: corrupt history file /home/kali/.zsh_history
(kali@thamim72~)
(kali@thamim72~) nano wf.c
(kali@thamim72~) gcc wf.c -o wf
(kali@thamim72~) ./wf
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
2
2
Enter Process Sizes:
22
2

Process No      Process Size    Block No
1                22             Not Allocated
2                2              1

(kali@YUVARAJK)-[~]
$
```

SHELL PROGRAMMING : WORST FIT

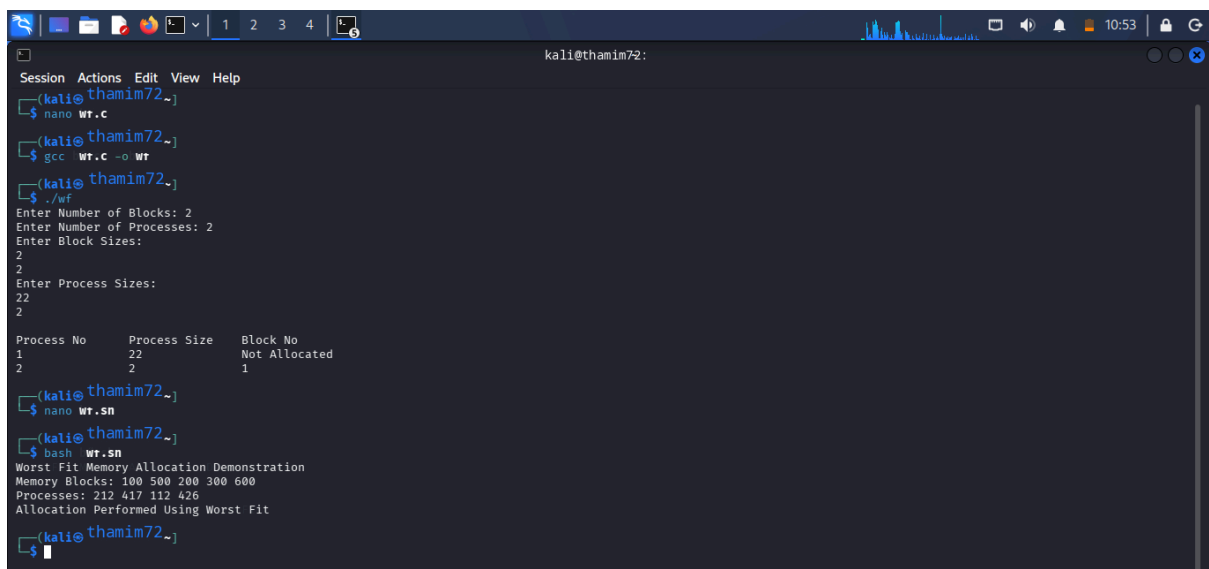


```
GNU nano 9.0 wf.sh
#!/bin/bash
echo "Worst Fit Memory Allocation Demonstration"
blocks=(100 500 200 300 600)
processes=(212 417 112 426)
echo "Memory Blocks: ${blocks[@]}"
echo "Processes: ${processes[@]}"
echo "Allocation Performed Using Worst Fit"

# Enter Number of Blocks: 2
# Enter Block Sizes:
#
# Enter Process Sizes:
#
# Process No.    Process Size    Block No.
# -----
# 1             212             Not Allocated
# 2             417             Not Allocated
# 3             112             Not Allocated
# 4             426             Not Allocated

# End of Worst Fit Memory Allocation Demonstration
echo "Memory Blocks: 100 500 200 300 600"
echo "Processes: 212 417 112 426"
echo "Allocation Performed Using Worst Fit"
```

OUTPUT :



```
kali@thamim72:~$ nano wt.c
kali@thamim72:~$ gcc wt.c -o wf
kali@thamim72:~$ ./wf
Enter Number of Blocks: 2
Enter Number of Processes: 2
Enter Block Sizes:
2
2
Enter Process Sizes:
22
22
2

Process No      Process Size    Block No
-----
1               22             Not Allocated
2               22              1

kali@thamim72:~$ nano wt.sh
kali@thamim72:~$ bash wt.sh
Worst Fit Memory Allocation Demonstration
Memory Blocks: 100 500 200 300 600
Processes: 212 417 112 426
Allocation Performed Using Worst Fit
kali@thamim72:~$
```